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The Missing Profits of Nations

Contents

1 Big picture

2 Data and measurement (key objects)

- 2.1 Main data sources
- 2.2 Profitability: the paper’s key object
- 2.3 Foreign firms versus local firms
- 2.4 Definition of tax havens
- 2.5 Global summary statistics: what the database contains
- 2.6 Bilateral allocation variables

3 Models and empirical specifications (all equations + interpretation)

- 3.1 A micro benchmark from the literature
- 3.2 Baseline “excess profitability” estimator
- 3.3 Production-function rationale
- 3.4 Validation regression using U.S. multinationals
- 3.5 Allocation to source countries
- 3.6 Allocation to parent countries
- 3.7 Immediate versus ultimate destination
- 3.8 Correcting macroeconomic statistics

4 Identification and credibility

- 4.1 Why the baseline comparison is informative
- 4.2 Baseline assumptions
- 4.3 Placebo-style evidence from non-haven countries
- 4.4 Validation with detailed U.S. data

4.5	Capital intensity explains only a small part	
4.6	How large could true profitability differences be?	
4.7	Alternative methodology	
4.8	Bounds and uncertainty	
4.9	Remaining limitations	
5	Tables and figures	
5.1	Figure 2: excess profitability versus effective tax rates	
5.2	Figure 3: profits-to-wage ratios of U.S. affiliates over time	
5.3	Table 1: global accounting totals	
5.4	Figure 4: profitability of foreign versus local firms	
5.5	Table 2: shifted profits by haven and robustness	
5.6	Figure 5: can capital intensity explain haven profitability?	
5.7	Table 3: profit and tax losses in source countries	
5.8	Figures 6 and 7: who loses the profits, and who ultimately gets them?	
5.9	Table 4 and Figure 8: corrected macro data and factor shares	
6	Short summary	
7	Conclusion	

1 Big picture

- **Question.** How much profit do multinational companies shift to tax havens globally, which countries lose these profits, and how much does profit shifting distort standard macro statistics such as GDP, trade balances, and factor shares?
- **Core empirical idea.** The paper exploits a new macro fact from foreign affiliates statistics: in tax havens, foreign-owned firms are vastly more profitable than local firms, while in high-tax countries foreign firms are slightly *less* profitable than local firms.
- **Data innovation.** The paper combines national accounts, inward foreign affiliates statistics, detailed U.S. outward foreign affiliates statistics, and bilateral balance of payments data on service trade, direct investment income, and intra-group interest flows.
- **Baseline estimator.** The profitability of local firms in each haven is used as a counterfactual benchmark for what foreign-firm profitability would look like in the absence of profit shifting. Excess profitability of foreign firms in havens is interpreted as shifted profit.
- **Main result.** In the preferred estimate, **\$616 billion** of profit was shifted to tax havens in 2015. This is **36% of multinational profits** booked outside parent countries.
- **Heterogeneity across multinationals.** U.S. multinationals shift substantially more than others. The paper estimates that U.S. firms book **54%** of their foreign profits in havens, versus about **27%** for other multinationals.
- **Heterogeneity across countries.** High-tax European Union countries are the biggest losers. After reallocating shifted profits back to source countries, domestic profits would rise by about **20%** in high-tax EU countries, **10%** in the U.S., and about **5%** in developing countries. In tax havens, recorded profits would fall by about **55%**.
- **Which havens matter most?** Ireland is the largest single destination in the baseline estimates, followed by Caribbean havens, Singapore, the Netherlands, Switzerland, and Luxembourg.
- **Credibility result.** An entirely independent “excess cross-border transactions” method delivers a very similar global total, **\$646 billion**, reinforcing the benchmark estimate.
- **Macro implication.** Profit shifting makes non-haven countries look less profitable than they really are and makes havens look extraordinarily productive. Correcting for this effect raises measured capital shares in high-tax Europe and weakens the view that the rise of the capital share is only a North American phenomenon.
- **Takeaway.** The paper’s contribution is not just another estimate of tax avoidance. It provides a globally consistent accounting framework for reallocating profits across countries and shows that profit shifting materially changes the macro picture of globalization.

2 Data and measurement (key objects)

2.1 Main data sources

The paper builds a global macro database by combining four main ingredients.

- **National accounts.** These provide GDP, value added, compensation of employees, operating surplus, corporate profits, depreciation, trade balances, and related aggregates for the corporate sector.
- **Foreign affiliates statistics (FATS).** These statistics record the wages and profits of firms controlled by foreign owners. The core distinction in the paper is between **foreign firms** and **local firms** within each country.
- **U.S. outward FATS from the BEA.** These are especially detailed and are used for validation because they report country, industry, tangible capital, R&D, and profits of the foreign affiliates of U.S. multinationals over a long time period.
- **Bilateral balance of payments data.** These data are used to allocate shifted profits across countries, especially through high-risk service exports and intra-group interest flows.

The benchmark year is **2015**, although the paper also studies long-run dynamics using earlier U.S. data and selected European time series.

2.2 Profitability: the paper's key object

The central statistic is the **profit-to-wage ratio**. Let corporate output be

$$Y = F(K, AL) = rK + wL, \quad (1)$$

where K is capital, L labour, A labour efficiency, w the wage rate, and rK operating surplus net of depreciation.

Define the capital share as

$$\alpha = \frac{rK}{Y}. \quad (2)$$

If corporations pay net interest amounting to a fraction p of operating surplus, then pre-tax corporate profits are

$$\Pi = (1 - p) rK. \quad (3)$$

The paper defines profitability as

$$\pi = \frac{\Pi}{wL} = \frac{(1 - p)\alpha}{1 - \alpha}. \quad (4)$$

Interpretation. This is the amount of pre-tax profit booked per dollar of labour compensation. It is the main statistic used to compare foreign and local firms within countries.

2.3 Foreign firms versus local firms

For any country:

- **Foreign firms** are firms more than 50% owned by foreign shareholders.
- **Local firms** are all other resident firms, including purely domestic firms and domestic multinationals headquartered in that country.

The foreign-firm profitability ratio is

$$\pi_f = \frac{(1 - p_f)\alpha_f}{1 - \alpha_f}, \quad (5)$$

and the local-firm profitability ratio is defined analogously as π_l .

At the country level,

$$\pi = s\pi_f + (1 - s)\pi_l, \quad (6)$$

where s is the wage share paid by foreign firms.

Interpretation. The paper’s basic comparison is *within-country*: how much more profitable are foreign firms than local firms in the same haven?

2.4 Definition of tax havens

The paper classifies **41 countries and territories** as tax havens: five OECD countries or economies (Belgium, Ireland, Luxembourg, the Netherlands, and Switzerland) plus 36 non-OECD havens and territories. Puerto Rico is treated as a haven separate from the U.S. because it is outside the U.S. federal corporate income tax system and because profit shifting there affects measured U.S. GDP and factor shares.

2.5 Global summary statistics: what the database contains

Table 1 summarizes the global corporate accounts in 2015.

Global aggregate	Billions of current US\$
Global gross output (GDP)	75,039
Depreciation	11,940
Net output	63,099
Net corporate output	34,084
Net corporate profits	11,515
Net profits of foreign-controlled corporations	1,703
Of which shifted to tax havens	616
Net profits of local corporations	9,812
Corporate income taxes paid	2,154

Two summary facts are especially useful:

- multinational profits are about **15%** of global net corporate profits;
- the paper estimates that about **36%** of those multinational profits were shifted to havens.

2.6 Bilateral allocation variables

To allocate shifted profit across countries, the paper constructs:

- **high-risk service exports**, especially royalties, intellectual-property services, management services, ICT services, and financial services;
- **intra-group interest flows**;
- bilateral direct investment income flows used to trace where haven profits ultimately accrue.

The paper also corrects known data gaps by replacing incomplete haven inward statistics with mirror outward statistics from counterpart countries when necessary.

3 Models and empirical specifications (all equations + interpretation)

3.1 A micro benchmark from the literature

To motivate the macro approach, the paper discusses the standard micro-style regression:

$$\log(\pi_{ic}) = \alpha + \beta(\tau_p - \tau_c) + \delta Firm_i + \gamma Country_c + \varepsilon_{ic}, \quad (7)$$

where π_{ic} is pre-tax profits of firm i in country c , τ_c is the host-country tax rate, and τ_p is the parent-country tax rate.

Interpretation. In the micro literature, the coefficient on tax differentials captures how much reported profit responds to tax incentives. The paper’s argument is that a macro framework is needed to obtain globally consistent estimates by source country, haven, and parent country.

3.2 Baseline “excess profitability” estimator

For haven h , recorded foreign-firm profit is

$$\Pi_{f,h}^{obs} = \pi_{f,h} W_{f,h}, \quad (8)$$

where $W_{f,h}$ is the wage bill of foreign firms in haven h .

The baseline counterfactual assumes that absent shifting, foreign firms would have the same profitability as local firms:

$$\Pi_{f,h}^{cf} = \pi_{l,h} W_{f,h}. \quad (9)$$

Hence the baseline estimate of profits shifted into haven h is

$$S_h^{base} = \Pi_{f,h}^{obs} - \Pi_{f,h}^{cf} = (\pi_{f,h} - \pi_{l,h}) W_{f,h}. \quad (10)$$

Interpretation. If foreign firms in Ireland pay wages of $W_{f,h}$ and book much more profit per dollar of wage than local Irish firms, the gap is treated as inward-shifted profit.

3.3 Production-function rationale

The paper also writes foreign-firm profitability as

$$\pi_f = \left(\frac{K_f}{wL_f} \right) \cdot MPK_f \cdot (1 + \mu_f), \quad (11)$$

where K_f is capital, MPK_f is the marginal product of capital, and μ_f captures rents or markups.

Interpretation. This decomposition is useful because it shows what could make foreign firms truly more profitable than local firms: more capital intensity, different technology, or different product-market power. The whole credibility section of the paper is about showing that these channels are far too small to explain the enormous haven profitability gaps.

3.4 Validation regression using U.S. multinationals

Using the detailed outward FATS of U.S. multinationals, the paper estimates

$$\pi_{cti} = \alpha_t + \beta_{1t} \ln(K_{cti}) + \beta_{2t} \ln(RD_{cti}) + \gamma_t X_{ct} + \delta_t Haven_c + \varepsilon_{cti}, \quad (12)$$

where:

- π_{cti} is the profits-to-wage ratio of U.S. affiliates in country c , year t , industry i ;
- K_{cti} is tangible capital;
- RD_{cti} is R&D spending;
- X_{ct} includes GDP and population controls for host countries;
- $Haven_c$ is an indicator for the paper's haven list.

Interpretation. The coefficient δ_t measures whether U.S. haven affiliates remain exceptionally profitable after controlling for capital intensity, R&D, industry composition, and country characteristics.

3.5 Allocation to source countries

Let $\bar{x}_{ij}$ denote the excess high-risk service exports from haven i to source country j , and let $\bar{g}_{ij}$ denote excess intra-group interest receipts. The paper defines excess cross-border transactions as

$$t_{ij} = \bar{x}_{ij} + \bar{g}_{ij}. \quad (13)$$

The profits shifted into haven i are then allocated back to source countries proportionally:

$$S_{ij}^{source} = S_i \cdot \frac{t_{ij}}{\sum_j t_{ij}}. \quad (14)$$

Interpretation. If Luxembourg receives a disproportionate amount of high-risk royalties and interest from France, Germany, and Italy, then a correspondingly larger fraction of Luxembourg's shifted profits is reallocated back to those countries.

3.6 Allocation to parent countries

The paper also allocates shifted profits to the countries where ultimate parents are headquartered. In reduced form, if d_{ik} denotes the bilateral direct-investment-income connection between haven i and parent country k , then

$$S_{ik}^{parent} = S_i \cdot \frac{d_{ik}}{\sum_k d_{ik}}. \quad (15)$$

Interpretation. This gives a “who ultimately owns the shifted profits?” perspective, which is distinct from the “where were the profits made?” perspective.

3.7 Immediate versus ultimate destination

To move from immediate recipient havens to ultimate recipient havens, the paper uses a reallocation matrix Ω :

$$b = \Omega a, \quad (16)$$

where a is the vector of profits shifted on an immediate-counterpart basis and b is the vector on an ultimate-destination basis.

Interpretation. This matters because some EU havens appear to act as conduits. Profits may first go to Ireland or the Netherlands but ultimately end up in non-EU offshore havens.

3.8 Correcting macroeconomic statistics

The paper distinguishes between:

- profit shifting through transfer pricing and intangible relocation (**85%** of the total), and
- profit shifting through intra-group interest (**15%** of the total).

Operationally, for source country c , if S_c^{TP} is the transfer-pricing/intangible component, the corrected operating surplus and corporate value added are

$$(rK)_c^{corr} = (rK)_c + S_c^{TP}, \quad (17)$$

$$Y_c^{corr} = Y_c + S_c^{TP}, \quad (18)$$

so the corrected capital share is

$$\alpha_c^{corr} = \frac{(rK)_c^{corr}}{Y_c^{corr}}. \quad (19)$$

Interpretation. Profit shifting does not merely move tax revenue; it changes measured output, measured profits, and thus measured factor shares.

4 Identification and credibility

4.1 Why the baseline comparison is informative

The identifying idea is simple: compare foreign and local firms *inside the same haven*. This absorbs much of the country-level environment — wage levels, local institutions, accounting rules, and the general profitability of operating in that place. If foreign firms in Ireland or Singapore are dramatically more profitable than local firms in those same economies, the natural suspect is profit shifting.

4.2 Baseline assumptions

The benchmark interpretation requires three assumptions.

1. **Elasticity of substitution $\sigma = 1$.** With a Cobb–Douglas production structure, differences in capital intensity do not by themselves create profitability gaps between foreign and local firms.
2. **Same competitive environment.** Foreign and local firms within a haven are assumed to face the same degree of product-market and factor-market competition.
3. **No large inward shifting in the local sector.** The profitability of local haven firms is assumed not to be strongly inflated by shifting into domestically headquartered multinationals.

The paper does not simply assert these assumptions. It stress-tests all of them.

4.3 Placebo-style evidence from non-haven countries

A key fact is that in high-tax countries foreign firms are not unusually profitable. If anything, they are slightly *less* profitable than local firms. For example, in the United Kingdom the profits-to-wage ratio is about **26%** for foreign firms versus **48%** for local firms.

Why this matters. If the foreign-local profitability gap were driven by generic multinational advantages, we would expect foreign firms to dominate local firms everywhere. Instead, the giant profitability gaps are highly concentrated in tax havens.

4.4 Validation with detailed U.S. data

The U.S. outward FATS provide the cleanest validation exercise. Haven affiliates of U.S. multinationals had a profits-to-wage ratio of about **346%** in 2015, compared with only **46%** for non-haven affiliates. The gap remains very large after controlling for:

- tangible capital,
- R&D spending,
- industry composition,
- host-country GDP and population.

Why this matters. The gap is not just that tax-haven affiliates are in different industries or use more capital or do more R&D.

4.5 Capital intensity explains only a small part

The paper explicitly checks whether foreign firms in havens are simply more capital intensive. This matters because if $\sigma \neq 1$, capital intensity can mechanically generate some profitability differences.

The quantitative result is modest:

- if $\sigma = 1.3$, the global shifted-profit estimate falls only from **616** to **586** billion dollars;
- if $\sigma = 0.7$, it rises to **663** billion dollars.

Interpretation. Even when using plausible non-unit elasticities, capital intensity explains little of the foreign-local profitability gap in havens.

4.6 How large could true profitability differences be?

The paper also allows foreign haven firms to be genuinely more profitable than local firms for reasons unrelated to tax avoidance. Using outside evidence on foreign ownership, firm size, and productivity, it increases the counterfactual profitability benchmark by **45 percentage points**. Even then, the global estimate only falls from **616** to **506** billion dollars.

Interpretation. To erase the haven profitability gap, foreign firms would need to be implausibly more productive than local firms.

4.7 Alternative methodology

A fully independent method based on excess high-risk service exports and intra-group interest receipts yields a global estimate of **646** billion dollars, very close to the baseline **616**.

Why this matters. Similar magnitudes emerge from two different data systems:

- foreign-vs-local profitability in FATS;
- abnormal cross-border transactions in the balance of payments.

4.8 Bounds and uncertainty

Table 2 reports a range of robustness checks and summary bounds.

Scenario	Global shifted profits (US\$ bn)	Share of multinational profits
Baseline	616	36%
Alternative transactions method	646	slightly above baseline
Higher benchmark profitability	506	lower
$\sigma = 1.3$	586	lower
$\sigma = 0.7$	663	higher
Allow inward shifting in local haven sector	693	higher
Remove data corrections	507	lower
Lower bound	369	22%
Upper bound	740	43%

Bottom line. The exact number is uncertain, but the conclusion that global profit shifting is quantitatively large is very robust.

4.9 Remaining limitations

The main limitations are measurement, not conceptual confusion:

- some havens report incomplete inward investment statistics;
- bilateral allocations are noisy because multinational structures are complex;
- the paper is an accounting exercise, not a structural general-equilibrium model of how firms would reorganize under a new tax system.

That said, the paper’s contribution is precisely to improve the accounting baseline that any structural policy model would need.

5 Tables and figures

5.1 Figure 2: excess profitability versus effective tax rates

Read. The graph plots $(\pi_f - \pi_l)$ against the effective corporate income tax rate. Tax havens cluster in the lower-left tax-rate region and have large positive profitability gaps. Non-haven countries mostly have gaps close to zero or negative.

Economic message. Abnormal profitability is not randomly distributed across countries. It lines up strongly with low-tax jurisdictions, exactly as a profit-shifting story would predict.

5.2 Figure 3: profits-to-wage ratios of U.S. affiliates over time

Read. In the 1960s and 1970s, haven and non-haven affiliates of U.S. multinationals had similar profitability. By 2015, haven affiliates had a profits-to-wage ratio of about **346%**, compared with **46%** for non-haven affiliates. The haven premium remains after controlling for capital, R&D, industry, and country characteristics.

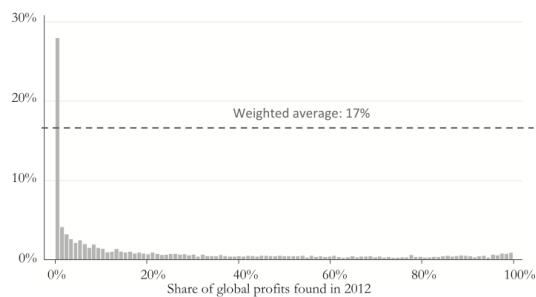


FIGURE 1
Fraction of global profits that can be observed in Orbis

Notes: This graph shows the density of the following ratio for the year 2012. For each multinational firm in Orbis, we compute the sum of the unconsolidated pre-tax profits of all subsidiaries (code U1), and we divide this sum by the consolidated global profits of the firm (code C1). Whenever the ratio is less than 1, this means that only part of the global profits of the firm are visible at the subsidiary level in Orbis. In 28% of the cases, no profits are visible at the subsidiary level. The weighted average of 17% is weighed by profit. Source: authors' computations using Orbis data.

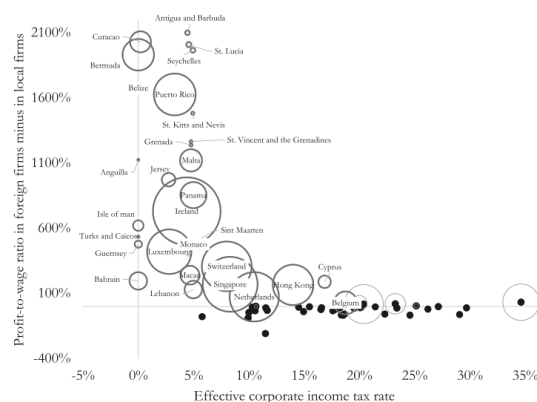


FIGURE 2
Excess profitability vs. effective tax rates

Notes: This graph shows the difference between the profits-to-wage ratio of foreign firms (π_f) and local firm (π_l), plotted against the effective corporate income tax rate. The year is 2015. The sample includes OECD countries, a number of developing non-OECD countries (Brazil, China, Colombia, Costa Rica, India, Russia, and South Africa), and the countries and territories included in our list of tax havens. Bubble sizes are proportional to the excess of π_f over π_l . In red are the countries and territories included in our list of tax havens. In blue are the countries not included in our list of havens that have π_f larger than π_l (with bubble size proportional to the excess of π_f over π_l). In black are all other non-haven countries (which have π_f lower than π_l). The effective corporate income tax rate is provided by the effective tax rate of the affiliates of US multinational companies, computed as foreign corporate income tax paid divided by profit-type return in the BEA survey of the foreign activities of US multinationals. Source: authors' computations.

Economic message. Profit shifting is a *rising* phenomenon. The dramatic time trend strongly suggests that the modern gap is not a timeless technological fact; it emerged together with the growth of multinational tax planning.

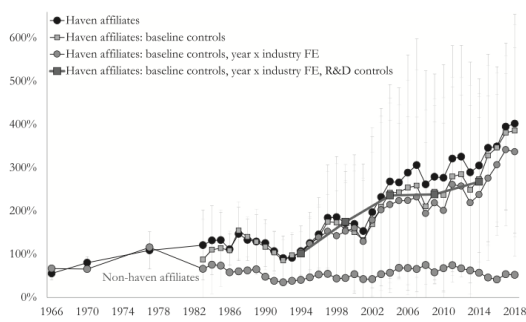


FIGURE 3
Profits-to-wage ratio of the foreign affiliates of US multinationals

Notes: This graph shows the profits-to-wage ratio of the affiliates of US multinationals in non-haven countries (blue line), in tax havens (black line), and in tax havens after controls, following equation (4). See text for description of each specification. R&D expenditures are linearly interpolated between benchmark surveys. Source: Authors' computations using tabulations of the BEA survey of the foreign operations of US multinationals. Profits are measured as profit-type return (Tables IL.F.1); see Supplementary Appendix Section A for complete details on variable construction. Regression coefficients for the profitability premium of haven affiliates δ_t are reported in Supplementary Appendix Figure A, with robust standard errors clustered at the country level. The 95% confidence intervals reported here are constructed using these standard errors.

TABLE 1
Global output, corporate output, and corporate taxes paid

	Billions of current US\$	% of net corporate profits
Global gross output (GDP)	75,039	
Depreciation	11,940	
Net output	63,099	
Net corporate output	34,084	296
Net corporate profits	11,515	100
Net profits of foreign-controlled corp.	1,703	15
Of which: shifted to tax havens	616	5
Net profits of local corporations	9,812	85
Corporate income taxes paid	2,154	19

Notes: This table reports the global totals in our database in 2015. Profits of foreign corporations include all the profits made by companies more than 50% owned by a foreign country; profits of local corporations equal all corporate profits minus the profits of foreign corporations. Source: Replication Guide Tables C.5 and A.3.

5.3 Table 1: global accounting totals

Read. The global totals imply:

Object	Value (US\$ bn)	Interpretation
Net corporate profits	11,515	total corporate profits worldwide
Multinational profits	1,703	profits made outside parent countries
Shifted to havens	616	about 36% of multinational profits
Corporate taxes paid	2,154	about 19% of net corporate profits

Economic message. The shifted-profit total is not a tiny accounting footnote. It is large relative

to multinational profits and large enough to affect country-level GDP, trade balances, and factor shares.

5.4 Figure 4: profitability of foreign versus local firms

Read. This is the paper's core descriptive figure. Foreign firms in tax havens are extraordinarily profitable:

- Ireland: around **800%** profit per dollar of wages,
- Puerto Rico: around **1,625%**,
- overall, foreign firms in havens: about **277%**,
- local firms in havens: about **34%**.

By contrast, in high-tax countries foreign firms are slightly less profitable than local firms.

Economic message. This one figure almost contains the whole paper. Foreign firms in havens look nothing like either local haven firms or foreign firms in non-havens.

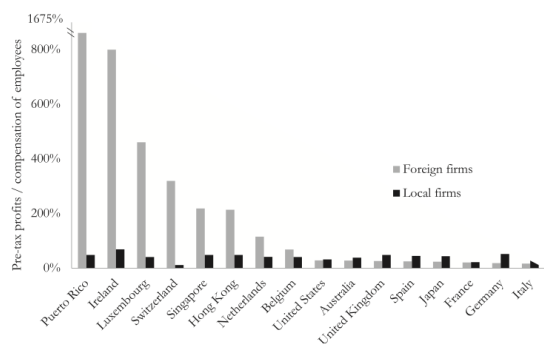


FIGURE 4
Profitability in foreign vs. local firms

Notes: This figure shows the ratio of pre-tax profits to compensation of employees for local firms (π_L) and foreign firms (π_F) in 2015, in the eight largest tax havens and the eight largest non-haven high-income countries in our sample. *Source:* Replication Guide Table A.7.

	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
	Baseline estimates (excess profitability)	Alternative estimate (excess transactions)	45 pp increase in benchmark profitability	Adjustment for capital intensity with $\sigma = 0.7$	Adjustment for capital intensity with $\sigma = 1.3$	Accounting for inward shifting headquarters	Removing corrections to haven data	Baseline estimates: lower bound	Baseline estimates: upper bound
Belgium	13	34	0	16	11	20	12	0	23
Ireland	106	89	100	115	102	109	95	84	118
Luxembourg	47	88	42	46	47	47	30	26	46
Malta	12	4	12	13	12	12	8	7	13
Netherlands	57	74	22	70	51	66	48	6	79
Caribbean	97	91	95	104	92	97	70	64	105
Bermuda	24	22	23	25	23	26	14	13	27
Singapore	70	74	52	88	61	75	48	20	93
Puerto Rico	42	44	41	42	41	42	42	40	42
Hong Kong	39	28	28	41	38	72	35	23	74
Switzerland	58	50	50	57	59	75	58	50	74
Other havens	51	47	42	45	49	52	47	36	46
Total	616	646	506	663	586	693	507	369	740

Notes: This table reports estimates for the amount of profit shifted into tax havens in 2015, in billions of US\$. The first column reports our baseline estimates, obtained by equating the profitability of foreign and local firms within each tax haven. Column 2 reports an alternative estimate based on the excess cross-border transactions conducive of profit shifting observed in the balances of payments of tax havens. Columns 3–7 shows how the baseline estimates from Column 1 are affected when changing one assumption at a time. Finally, Columns 8 and 9 report lower and upper bounds for our baseline estimates, obtained by changing all our assumptions at the same (either in the most conservative or least conservative manner).

5.5 Table 2: shifted profits by haven and robustness

Read. The main haven destinations in the baseline are:

Haven	Baseline	Lower bound	Upper bound
Ireland	106	84	118
Caribbean havens	97	64	105
Singapore	70	20	93
Switzerland	58	50	74
Netherlands	57	6	79
Luxembourg	47	26	46
Puerto Rico	42	40	42
Hong Kong	39	23	74
Total	616	369	740

The alternative transactions-based total is **646**.

Economic message. Profit shifting is concentrated in a small number of hubs, with Ireland playing the single biggest role in the point estimates. The basic pattern is robust even under conservative assumptions.

5.6 Figure 5: can capital intensity explain haven profitability?

Read. Foreign haven firms sometimes do use more capital than local firms, but the capital-intensity differences are tiny compared with the profitability differences. In Ireland, even if local firms were given foreign-firm capital intensity and one assumes $\sigma = 1.3$, local profitability would rise only to about **100%** — still far below the observed foreign-firm profitability of **800%**.

Economic message. Real capital location is not the main explanation. The bulk of excess profitability in havens reflects booked profit, not real capital deepening.

5.7 Table 3: profit and tax losses in source countries

Read. Selected country losses are:

Country	Shifted profits	% GDP	% reported profits	% tax collected
United States	143	0.8	8	14
Germany	55	1.6	10	28
United Kingdom	62	2.1	14	18
France	32	1.3	17	21
Italy	23	1.2	11	19
China	61	0.5	3	4
Brazil	14	0.6	5	9

The text also emphasizes regional totals:

- U.S.: **\$143** billion,

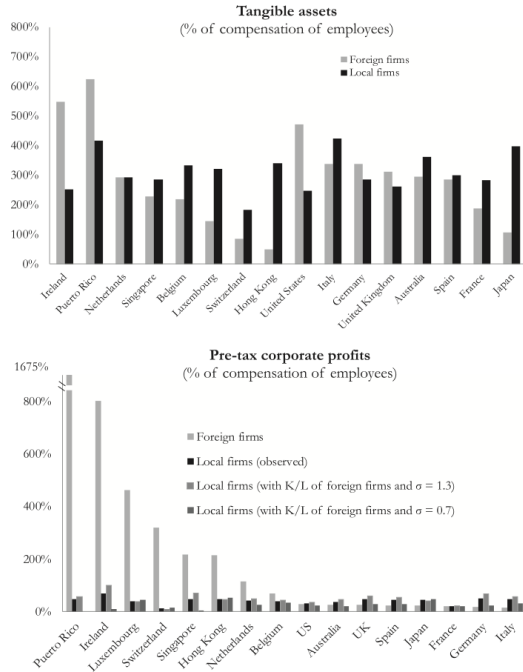


FIGURE 5
Capital intensities in tax havens

Notes: The top panel shows the ratio of the tangible capital stock of local and foreign firms to wages paid in 2015, in the eight largest tax havens and the eight largest non-haven high-income countries in our sample. The bottom panel shows, in the same sample, the profit-to-wage ratio in foreign firms and in local firms obtained by applying the capital intensity of foreign firms, for different assumptions about the elasticity of substitution between capital and labour σ . Source: Replication Guide Table E.1.

- EU high-tax countries: **\$216** billion,
- other OECD countries: **\$76** billion.

Economic message. The European Union is the main source region. The U.S. also loses a lot in absolute dollars, but Europe loses more relative to GDP.

5.8 Figures 6 and 7: who loses the profits, and who ultimately gets them?



FIGURE 6
Allocating the shifted profits

Notes: This figure shows the regions from which the profits shifted to tax havens originate (pink bar), and the regions where the ultimate parent companies to which the shifted profits ultimately accrue are incorporated (blue bar) in 2015. In both cases, profits are expressed as a fraction of the global amount of profits shifted to tax havens, \$616 billion in 2015. Source: Replication Guide Tables C.4b, C.4c, and C.4d.

TABLE 3
Profit and corporate income tax revenue losses

	[1]	[2]	[3]	[4]	[5]	[6]	[7]
	Reported domestic profits	Of which: local firms	Of which: foreign firms	Shifted profits (baseline)	Shifted profits (% reported GDP)	Shifted profits (% reported profits)	Corporate tax loss (% tax collected)
OECD countries							
Australia	179	151	28	13	1.1	7	7
Austria	48	37	11	4	0.9	7	11
Canada	143	96	47	19	1.2	13	10
Chile	68	58	10	5	2.3	8	12
Czech Republic	34	16	17	2	0.9	5	5
Denmark	52	47	5	3	1.0	6	8
Estonia	4	3	1	0	1.1	6	10
Finland	25	21	4	3	1.2	11	11
France	188	156	32	32	1.3	17	21
Germany	553	510	43	55	1.6	10	28
Greece	23	21	1	1	0.5	5	7
Hungary	21	11	10	2	2.0	12	21
Iceland	2	2	0	0	2.6	20	22
Ireland	54	48	6	1	0.2	1	2
Italy	212	199	13	23	1.2	11	19
Japan	634	602	32	9	0.2	1	2
Korea	248	246	3	5	0.3	2	3
Latvia	4	3	1	0	0.7	5	7
Mexico	325	302	23	13	1.1	4	11
New Zealand	44	37	6	2	0.9	3	5
Norway	76	69	7	5	1.3	7	8
Poland	88	68	19	4	0.8	4	8
Portugal	27	22	5	3	1.3	10	9
Slovakia	12	6	5	1	0.7	5	5
Slovenia	3	2	1	0	0.5	7	6
Spain	159	138	21	14	1.2	9	14
Sweden	63	39	24	9	1.7	13	13
Turkey	213	209	4	5	0.6	2	8
UK	425	353	72	62	2.1	14	18
US	1,889	1,737	153	143	0.8	8	14
Non-OECD countries							
Brazil	274	245	30	14	0.6	5	9
China	2,069	1,906	162	61	0.5	3	4
Colombia	59	52	7	1	0.5	2	2
Costa Rica	13	12	1	1	1.9	8	21
India	376	368	8	9	0.4	3	9
Russia	290	253	37	12	0.9	4	6
South Africa	76	68	9	4	1.2	5	6
Other non-havens	1,423	1,309	114	79	0.8	6	7
Non-haven total	11,515	9,812	1,703	616	0.8	5	9

Notes: This table shows pre-tax corporate profits in OECD countries and a number of non-OECD countries other than tax havens, as published in national account statistics. Column 4 reports our baseline estimate of the amount of profit shifted out of each of these countries, which is then expressed as a fraction of reported GDP (Column 5) and reported profits (Column 6). Column 7 shows the implied corporate income tax revenue loss, obtained by applying the statutory corporate income tax rate to the amount of shifted profits reported in Column 4. Amounts are in current billion US\$. All data are for 2015. Source: Replication Guide Tables A.6., A.7, C.4d, and C.4.

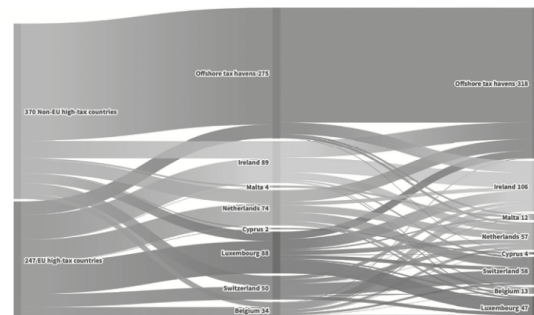


FIGURE 7
Mapping the ultimate destination of shifted profits

Notes: This diagram shows intragroup interest payments and exports conducive of profit shifting from payer/buyer countries (on the left) to immediate recipient havens (in the middle) and ultimate recipient havens (on the right) in 2015. The left panel visualizes profit shifting on an immediate counterpart basis (payments from high-tax countries to the immediate counterpart haven) while the right panel visualizes how havens transact among themselves. Offshore tax havens denote the aggregate of all non-EU havens. An interactive version of this diagram is available at <https://public.flourish.studio/story/939975/>.

Read. Figure 6 compares two allocations:

- **source-country allocation:** where the shifted profits were stripped from;
- **parent-country allocation:** where the parent firms that ultimately own those profits are located.

Two striking asymmetries appear:

- only about **23%** of globally shifted profits are shifted *out of* the U.S.;
- but about **50%** of globally shifted profits ultimately accrue to *U.S. parent companies*.

Figure 7 then shows that about **77%** of profits shifted out of the European Union initially go to EU havens, especially Ireland, Luxembourg, and the Netherlands, but about **half** of those profits ultimately end up in non-EU offshore havens.

Economic message. The U.S. is the main user of haven structures, while the EU is the main source region suffering profit losses. EU havens often act as conduits rather than final destinations.

5.9 Table 4 and Figure 8: corrected macro data and factor shares

TABLE 4
Macro statistics corrected for profit shifting

	Corrected capital share (%)	Difference with published data (%)	Corrected trade balance (% GDP)	Difference with published data (%)
OECD countries				
Australia	26	+1.2	-1.3	+0.9
Austria	29	+1.3	4.1	+0.8
Canada	24	+1.5	-1.4	+1.0
Chile	51	+1.6	2.0	+1.9
Czech Republic	39	+1.0	6.5	+0.7
Denmark	31	+1.2	8.2	+0.8
Estonia	35	+1.1	5.0	+0.9
Finland	28	+1.6	0.9	+1.0
France	19	+2.1	0.4	+1.1
Germany	31	+1.8	9.2	+1.2
Greece	43	+1.1	0.3	+0.5
Hungary	39	+2.3	10.4	+1.5
Iceland	33	+3.4	9.5	+2.0
Israel	37	+0.2	3.2	+0.2
Italy	30	+1.9	3.9	+1.0
Japan	27	+0.3	-0.4	+0.2
Korea	38	+0.4	8.0	+0.3
Latvia	31	+0.9	-0.5	+0.6
Mexico	71	+0.7	-1.1	+1.0
New Zealand	44	+0.7	1.5	+0.7
Norway	41	+1.3	6.5	+1.1
Poland	45	+0.9	3.7	+0.6
Portugal	33	+1.8	2.8	+1.1
Slovakia	35	+1.1	3.4	+0.6
Slovenia	18	+0.9	9.1	+0.4
Spain	29	+1.6	3.4	+1.0
Sweden	31	+2.2	6.3	+1.4
Turkey	55	+0.5	-2.3	+0.5
UK	31	+2.5	0.2	+1.8
US	27	+1.1	-2.1	+0.7
Non-OECD countries				
Brazil	26	+1.1	-0.2	+0.5
China	44	+0.6	3.7	+0.5
Colombia	55	+0.5	-5.9	+0.5
Costa Rica	45	+2.0	1.7	+1.8
India	56	+0.6	-2.5	+0.4
Russia	40	+0.9	8.9	+0.7
South Africa	39	+1.5	0.1	+1.1
Main havens				
Belgium	24	-2.5	0.1	-2.2
Ireland	42	-20.0	-5.8	-36.8
Luxembourg	27	-11.8	4.0	-30.0
Netherlands	32	-2.0	6.4	-4.2
Singapore	23	-7.1	3.9	-21.3
Puerto Rico	30	-37.8	-20.4	-43.5

Notes: This table shows corrected capital shares and trade balances in our sample of non-haven countries and in the main tax havens in 2015; see text for the construction of these corrections. Source: Replication Guide Tables C.5 and C.5b.

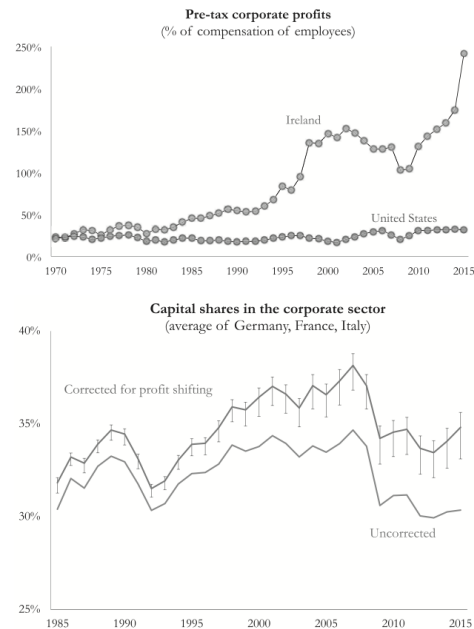


FIGURE 8
The Rise of Profit Shifting: Implications for Factor Shares

Notes: the top panel shows the ratio of profits to wages in Ireland and the US, using national accounts data. The bottom panel shows the arithmetic average of the gross-of-depreciation corporate capital share (gross corporate profits relative to gross corporate value-added) in Germany, France, and Italy. The blue line is taken from Gutiérrez and Pilon (2020) and the red line is adjusted for profit shifting; see text. The point estimate corresponds to our preferred estimate of global profit shifting and the confidence interval is constructed using the lower and upper bounds for global profit shifting reported in Table 2, Columns 8 and 9.

Read. Correcting for profit shifting changes capital shares and trade balances materially.

Country	Capital-share correction (pp)	Trade-balance correction (% GDP)
France	+2.1	+1.1
Germany	+1.8	+1.2
Italy	+1.9	+1.0
United States	+1.1	+0.7
Ireland	-20.0	-36.8
Luxembourg	-11.8	-30.0
Puerto Rico	-37.8	-43.5

Figure 8 then shows that for the average of Germany, France, and Italy, the corrected corporate capital share rises by about **3.0 percentage points** between 1985 and 2015 in the preferred scenario, and by about **3.5 points** in the upper-bound scenario, whereas the uncorrected series looks roughly flat.

Economic message. Profit shifting does not just change tax-revenue statistics; it changes the long-run macro narrative. Europe also experienced a rise in the capital share, but the official data hide part of it.

6 Short summary

This paper’s core contribution is a simple but powerful global accounting method for measuring profit shifting. The central empirical fact is that foreign firms in tax havens are dramatically more profitable than local firms in those same economies, whereas foreign firms in non-havens are slightly less profitable than local firms. Interpreting that excess haven profitability as shifted profit yields a global estimate of about **\$616 billion** in 2015, or **36%** of multinational profits.

The economic geography is asymmetric. U.S. multinationals appear to be the most aggressive shifters, but high-tax European countries are the main losers relative to GDP. Once profits are reallocated back to source countries, GDP, corporate profits, trade balances, and capital shares all change materially. The paper therefore matters for both public-finance questions about tax avoidance and macroeconomic questions about factor shares.

7 Conclusion

- **Core conclusion.** Global profit shifting is large, measurable with macro data, and concentrated in a small set of tax havens.
- **Economic intuition.** If a multinational can move paper profits to jurisdictions with very low tax rates while leaving workers and real activity elsewhere, haven affiliates will appear abnormally profitable and non-haven countries will look artificially unprofitable.
- **Most important cross-country result.** The European Union loses the most relative to GDP, while U.S. parent firms capture the largest share of the shifted profits.
- **Most important macro result.** Correcting for profit shifting raises measured capital shares in

non-haven countries and lowers them sharply in havens. This means official statistics understate the rise of capital's share in Europe.

- **Why the paper matters.** It gives researchers and policymakers a globally consistent way to talk about where profits are booked, where they were likely earned, and who ultimately owns them.
- **Takeaway.** The “missing profits” are not small. They are large enough to reshape how we think about corporate taxation, international redistribution, and even the measured evolution of capitalism across countries.